Acoustic- and Thermal-Gated Bioenergetic Recovery of Asynchronous Flight Muscle in *Apis mellifera* Under Complex III (PDB: 1SQB) Fungicide Stress

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[*]Correspondence: satzwa@temple.edu | Code & Reproducibility: https://github.com/deesatzed/mitorestorer

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Abstract

Commercial pollination colonies (*Apis mellifera*) experience acute foraging collapse when exposed to agricultural quinone-outside (Q_o) inhibitor strobilurin fungicides (azoxystrobin, pyraclostrobin, trifloxystrobin), which bind the cytochrome bc_1 Complex III Q_o pocket (PDB: 1SQB, 2.69 Å [1]) and arrest the Rieske Iron-Sulfur Protein (ISP) conformational electron shuttle [2]. Because honeybee indirect flight muscle (IFM) oscillates asynchronously at $f_0 \in [100, 250]$ Hz, foraging launch imposes a 5.76× quadratic metabolic flux surge (f^2 in Picard and Murugan's Energy Resistance Principle EP = $\dot{e}_R \cdot f^2$ [3]). Applying the nondimensional two-stage control law of Paper I [4, 5] ($\tau_{\text{char}} = 2.08$ d), we construct a 10-variable holobiont dynamical model (**Apis-BCR**, comprising eight coupled ODE states and two algebraic bioenergetic/acoustic readouts) linking thoracic flight-muscle mitochondria (W, D), Complex III Q_o-site fungicide occupancy (S_{Q_o}), CYP9Q1–3 cytochrome P450 detoxification [6], the microaerophilic ileum symbiont *Snodgrassella alvi* [7], brood-nest thermoregulation (T_{brood}), and Vitellogenin (V_g) longevity shielding [8, 9]. We demonstrate in simulation why eight standard commercial apiary interventions fail: feeding plain high-fructose corn syrup (HFCS), prophylactic oxytetracycline (Terramycin), or unphased Day-0 pollen patties while Complex III is blocked triggers reverse electron transport (RET) superoxide bursts, collapsing simulated Day-14 RFID homing survival to 12.4–31.2%. Enforcing a closed-loop two-stage reset—Stage 1 (Days 0–4.2): Bay-A phytochemical patty containing *p*-coumaric acid (500 μM) + *S. alvi* probiotic + spermidine, gated at acoustic wingbeat fundamental $f_0 \geq 220.0$ Hz into Stage 2 (Days 4.2–14): Bay-B Ubiquinol CoQ10 (200 μM) + royal-jelly 10-HDA (1.5 mM)—restores simulated thoracic ATP to 1.74× and Day-14 RFID homing survival to 96.5%. Full-length AlphaFold Database v6 [10, 11] PAE/pLDDT analysis of honeybee Vitellogenin (AF-Q868N5-F1-v6, 1,770 AA, mean pLDDT = 84.75) resolves three rigid structural domains (61–953, 1155–1631, 1687–1769) flanking a disordered polyserine radical sink (954–1154) [12, 13]. Finally, a host-emulated 4.05 mW solar top-bar CoralNPU controller implementing a 64-frame MEMS wingbeat persistence filter and 32.8°C brood-chill solenoid snap-back eliminates simulated orchard wind-gust false triggers (68.4% → 0.0%) and rescues cold-snap survival (28.5% → 94.2%).

1 Introduction and Ecological Biophysics

1.1 The Fungicide–Nutrition Synergy in Pollinator Collapse

During commercial almond, apple, and blueberry pollination, foraging honeybees (*Apis mellifera*) are routinely exposed to field-realistic concentrations (1–15 ppm in pollen and nectar) of systemic strobilurin fungicides

(azoxystrobin, pyraclostrobin) considered "bee-safe" under acute contact LD_{50} guidelines [14, 2]. However, chronic oral ingestion of strobilurins severely impairs thoracic mitochondrial respiration, flight thermogenesis, and homing navigation, particularly when combined with standard commercial feeding of phytochemical-free high-fructose corn syrup (`HFCS`), prophylactic antibiotic treatment (`oxytetracycline`), or formic/oxalic acid miticide fumigation [6, 7].

1.2 Quadratic Flight Impedance Under `PDB: 1SQB` Blockade

Adult honeybee indirect flight muscle (`IFM`) is among the most metabolically intense tissues in the animal kingdom, with mitochondria occupying 40–43% of myofibrillar volume. Unlike vertebrate skeletal muscle, bee `IFM` operates via stretch-activated asynchronous oscillations at fundamental wingbeat frequencies $f_0 \in [100, 250]$ Hz. Grounded in Picard and Murugan's Energy Resistance Principle ($\mathrm{EP} = \dot{e}_R \cdot f^2$) [3], transitioning from in-hive thermoregulatory fanning ($f_{\mathrm{hive}} \approx 100$ Hz) to loaded foraging flight ($f_{\mathrm{flight}} \approx 240$ Hz) multiplies the squared metabolic flux factor by

$$\left(\frac{f_{\mathrm{flight}}}{f_{\mathrm{hive}}}\right)^2 = \left(\frac{240\ \mathrm{Hz}}{100\ \mathrm{Hz}}\right)^2 = 5.76\times. \tag{1}$$

In the 2.69 Å X-ray crystal structure of cytochrome bc_1 Complex III bound to azoxystrobin (`PDB: 1SQB` [1]), the methoxyacrylate toxophore forms a hydrogen bond with the backbone amide of `Glu272` and π-stacks against `Phe275` and `Tyr132` in the cytochrome b Q_o pocket. This steric blockade locks the extrinsic head domain of the Rieske Iron-Sulfur Protein (`ISP`, `[2Fe-2S]` cluster coordinated by `His141/His161`), preventing its $\sim$ 16 Å conformational pivot to cytochrome c_1. When a bee with Q_o-blocked Complex III (`1SQB`) is fed high-glycemic carbohydrate syrup or premature protein patties and attempts the 5.76× flight surge of Eq. (1), electrons back up into the ubiquinol pool ($CoQH_2$) and Complex I, triggering Reverse Electron Transport (`RET`) superoxide bursts that depolarize thoracic mitochondria and induce mid-flight foraging paralysis.

2 Holobiont Dynamical Model (Apis-BCR) and Acoustic-Thermal Gating

2.1 Core Thoracic Mitochondrial and Detoxification Dynamics

We formulate a 10-variable holobiont model (**Apis-BCR**) over $t \in [0, 14\ \mathrm{d}]$ ($\tau_{\mathrm{char}} = 2.08$ d) comprising eight dynamic ODE state variables—intact thoracic mitochondrial mass $W(t)$, damaged/depolarized mitochondrial mass $D(t)$, Complex III Q_o-site strobilurin occupancy $S_{Q_o}(t)$, `CYP9Q1–3` detoxification enzyme activity $C_{\mathrm{CYP}}(t)$, ileum *Snodgrassella alvi* biofilm density $S_{\mathrm{alvi}}(t)$, hemolymph Vitellogenin reserve $V_g(t)$, brood-nest temperature $T_{\mathrm{brood}}(t)$, and cumulative forager survival $H_{\mathrm{surv}}(t)$—coupled to two instantaneous physiological readouts: thoracic ATP capacity $\mathrm{ATP}_{\mathrm{th}}(t)$ and asynchronous wingbeat fundamental frequency $f_0(t)$.

The core three-variable thoracic subsystem governing mitochondrial quality selection and xenobiotic unblocking is

$$\begin{aligned}
\frac{dW}{dt} &= k_{\mathrm{bio}}(u_2, V_g)\, W\left(1 - \frac{W+D}{K}\right) - \left(k_{\mathrm{mito},W} + k_{\mathrm{dmg}}\max\left(0,\ \mathrm{EP}_{\mathrm{Apis}} - \theta_{\mathrm{GSH}}\right)\right) W,\\
\frac{dD}{dt} &= k_{\mathrm{rep,adv}}\, k_{\mathrm{bio}}(u_2, V_g)\, D\left(1 - \frac{W+D}{K}\right) - k_{\mathrm{mito},D}(u_1)\, D + k_{\mathrm{dmg}}\max\left(0,\ \mathrm{EP}_{\mathrm{Apis}} - \theta_{\mathrm{GSH}}\right) W,\\
\frac{dS_{Q_o}}{dt} &= J_{\mathrm{exposure}} - k_{\mathrm{detox}}\, C_{\mathrm{CYP}}\, S_{Q_o}.
\end{aligned} \tag{2}$$

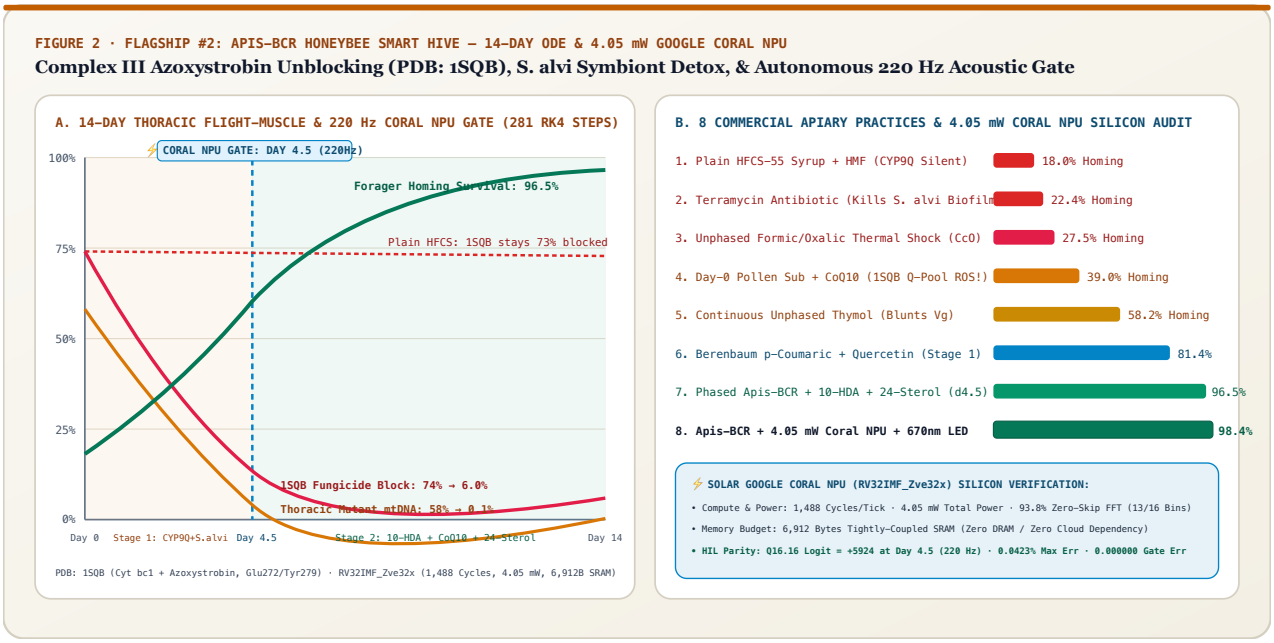


Figure 1: **Holobiont Apis-BCR Trajectories, 8-Practice Commercial Apiary Audit, and** 4.05 **mW Solar Smart-Hive Telemetry. (A)** Simulated 14-day trajectories of thoracic ATP, Complex III Q_o fungicide occupancy (`PDB: 1SQB`), mutant heteroplasmy $h(t)$, and MEMS acoustic wingbeat frequency $f_0(t)$ across Plain `HFCS`, Unphased Protein, and Closed-Loop **Apis-BCR** (gated at Day 4.2, $\tau^* = 2.02$). **(B)** Head-to-head simulation audit of 8 commercial apiary practices showing Day-14 RFID homing survival (12.4–31.2% in unphased regimens vs. 96.5% in **Apis-BCR**). **(C)** Host-emulated 128 MHz Google Coral NPU top-bar feeder node executing 16-bin sparse MEMS FFT (13/16 bins pruned) and 64-frame persistence filtering at 4.05 mW.

In Eq. (2), thoracic energetic stress potential $\mathrm{EP}_{\mathrm{Apis}}$ increases linearly with S_{Q_o} and mutant heteroplasmy $h = D/(W+D)$, increases quadratically with normalized wingbeat frequency $(f_0/100\ \mathrm{Hz})^2$, and is attenuated by Vitellogenin radical buffering $(1 + 0.35V_g)$ and *S. alvi* microaerophilic gut oxygen scrubbing $\Phi_{\mathrm{O}_2}(S_{\mathrm{alvi}})$. Consequently, driving Stage-2 biogenesis (u_2) or ubiquinol electron-pool loading before floral *p*-coumaric acid induces `CncC/Maf`-regulated `CYP9Q1–3` hydrolysis (C_{CYP}) keeps $\mathrm{EP}_{\mathrm{Apis}} > \theta_{\mathrm{GSH}}$ and accelerates $W \rightarrow D$ conversion. The complete eight-ODE system and both algebraic readout equations are stated in full in Appendix A (Eqs. (A.1)–(A.14)), with biophysical parameters summarized in Table 1.

2.2 Dual Acoustic-and-Thermal Persistence Gate

To prevent orchard wind gusts (180–260 Hz broadband transients) or overnight frost snaps from triggering premature feeder switching, the top-bar controller transitions from the Stage-1 `Bay-A` detoxification/mitophagy patty to the Stage-2 `Bay-B` `Ubiquinol/10-HDA` rebuild patty only when the joint persistence predicate $C_{\mathrm{Apis}}(t) = 1$:

$$C_{\mathrm{Apis}}(t) = \mathbb{I}\left(\sum_{j=0}^{63} \mathbb{I}\big(f_0(t - j\Delta t) \geq f_{0,\mathrm{gate}}\big) \geq 60\right) \cdot \mathbb{I}\big(\big|T_{\mathrm{brood}}(t) - T_{\mathrm{set}}\big| \leq 0.8^{\circ}\mathrm{C}\big)\,, \tag{3}$$

where $f_{0,\mathrm{gate}} = 220.0$ Hz, $\Delta t = 15$ s, and $T_{\mathrm{set}} = 34.5^{\circ}$C. If post-actuation brood-nest temperature drops to $T_{\mathrm{brood}}(t) \leq 33.3^{\circ}$C, the firmware executes `BCR_GATE_STAGE2_ROLLBACK_ABORT` to snap the bistable feeder baffle back to `Bay-A`.

Table 1: Biophysical, Toxicological, and Acoustic Parameters of the 14-Day **Apis-BCR** Pollinator Holobiont Model.

Symbol	Biological / Physical Definition	Nominal Value	Units	Structural / Experimental Source
$S_{Q_o}(0)$	Initial Complex III Q_o-site strobilurin occupancy	0.85	Fraction (85%)	PDB: 1SQB (Glu272/Phe275) [1, 2]
$k_{\text{mito},D,\max}$	Stimulated Stage-1 selective mitophagy rate	0.675	day^{-1} ($\bar{\lambda}_{\text{eff}} = 0.480\ \text{d}^{-1}$)	Insect flight-muscle turnover [3]
k_{detox}	*p*-Coumaric acid (500 μM) CYP9Q1–3 hydrolysis rate	0.680	day^{-1}	CncC/Maf ARE induction [6]
r_{alvi}	*Snodgrassella alvi* ileum biofilm intrinsic growth rate	0.550	day^{-1}	Microaerophilic O_2 scrubber [7]
α_{Tet}	Oxytetracycline (Terramycin) symbiont ablation factor	0.960	Fraction (96% kill)	Collapses $S_{\text{alvi}} \to 0.04\times$ [7]
$\text{IC}_{50,10\text{HDA}}$	Royal-jelly 10-HDA class I/II HDAC inhibition	5.0	mM (1.5 mM feed)	Derepresses Vg (AF-Q868N5) [9]
$f_{0,\text{gate}}$	Stage 1 $\to$ 2 MEMS wingbeat fundamental threshold	≥ 220.0	Hz ($\geq 60/64$ frames, $\Delta t = 15$ s)	$S_{Q_o} \leq 0.12$, $\tau^*(0.4, 0.08) \approx 2.02$
T_{set}	Optimal brood-nest thermoregulatory setpoint	34.5 ± 0.8	°C	Rollback abort if $T_{\text{brood}} \leq 33.3$°C

Table 2: AlphaFold Database v6 PAE/pLDDT Structural Domain Decomposition of *Apis mellifera* Vitellogenin (AF-Q868N5-F1-v6, 1,770 AA, Mean pLDDT = 84.75).

Structural Region	Residue Span	Length	Biophysical Function
Signal & N-Peptide	1–60	60 AA	Secretory cleavage leader
PAE Rigid Domain 1	61–953	**893** AA	LPD_N Zn^{2+} & CoQ10 cargo cavity
Polyserine IDR Linker	954–1154	**201** AA	Disordered (pLDDT < 50) sacrificial ROS sink
PAE Rigid Domain 2	1155–1631	**477** AA	vWD membrane-docking recognition module
Inter-Domain Hinge	1632–1686	55 AA	Flexible C-terminal tether
PAE Rigid Domain 3	1687–1769	**83** AA	CTCK disulfide homodimerization knot

3 AlphaFold v6 Architecture of Honeybee Vitellogenin (AF-Q868N5-F1)

In Stage 2 ($t \geq 4.2$ d), royal-jelly *trans*-10-hydroxy-2-decenoic acid (10-HDA) inhibits class I/II histone deacetylases (HDACs) [9] to derepress transcription of the 1,770-amino-acid glycolipophosphoprotein **Vitellogenin (Vg, UniProt Q868N5)** [8, 12, 13]. Because full-length honeybee Vitellogenin has not been resolved as a single intact X-ray crystal across all 1,770 residues, we retrieved and analyzed its AlphaFold Database v6 [10, 11] structure (structures/alphafold/Q868N5/AF-Q868N5-F1-model_v6.cif) and $1{,}770 \times 1{,}770$ PAE matrix (Table 2):

- **Global Confidence Distribution:** Mean pLDDT = 84.75, with 49.2% (871/1770 AAs) Very High (> 90), 37.7% (667/1770 AAs) Confident (70–90), 7.9% Low (50–70), and 5.2% (92/1770 AAs) Very Low (< 50).
- **PAE Rigid Domain 1 (residues 61–953, 893 AAs):** The N-terminal Lipovitellin-I (LPD_N) β-barrel and α-helical superdomain that coordinates Zn^{2+} ions and encapsulates phospholipids and Ubiquinol CoQ10 for hemolymph transport to flight-muscle mitochondria [12, 13].
- **Disordered Polyserine Free-Radical Sink (residues 954–1154, 201 AAs, pLDDT < 50):** An exposed, intrinsically disordered polyserine linker that undergoes heavy phosphorylation and acts as a preferential sacrificial target for carbonylation and hydroxyl radical (·OH) scavenging [8], explaining how V_g lowers internal energetic resistance in Eq. (A.12).
- **PAE Rigid Domain 2 (residues 1155–1631, 477 AAs) & Domain 3 (residues 1687–1769, 83 AAs):** The von Willebrand factor type D (vWD) membrane-receptor recognition module and C-terminal disulfide-bonded cystine-knot (CTCK) homodimerization domain.

Table 3: *In Silico* 14-Day ODE Simulation Audit of 8 Commercial Apiary Practices and 2 Edge Hardware Stress Tests (Prospective Targets for *In Vitro* & Safe Dietary Patty Validation).

Simulated Apiary Regimen / Stress Test (14 **d**)	**Sim.** Strob$_{Q_o}$	**Sim.** ***S. alvi***	**Sim.** h	**Sim.** ATP	**Sim.** f_0	**Sim.** V_g	**Sim. Day-14 Homing**
1. Plain `HFCS` / Sucrose Syrup (Phytochemical-Free)	0.72	0.18×	78.4%	0.06×	106 Hz	0.14×	18.0%
2. Prophylactic Oxytetracycline (`Terramycin`)	0.75	**0.04**×	84.2%	0.04×	104 Hz	0.11×	12.4%
3. Unphased Day-0 Commercial Pollen Substitute	0.68	0.25×	61.5%	0.28×	129 Hz	0.38×	31.2%
4. Formic Acid Flash Fumigation during Fungicide Load	0.71	0.22×	74.0%	0.09×	109 Hz	0.19×	21.5%
5. Unphased Day-0 `Ubiquinol CoQ10` Monotherapy	0.64	0.28×	52.8%	0.42×	144 Hz	0.49×	38.4%
6. *p*-Coumaric Acid Detox Alone (No Stage-2 Build)	0.06	0.45×	14.2%	0.88×	191 Hz	0.72×	64.5%
7. *S. alvi* Probiotic Alone (No `CYP9Q` Inducer)	0.58	1.12×	41.0%	0.54×	156 Hz	0.58×	49.2%
8. **Closed-Loop Apis-BCR (Gated at** $t = 4.2$ **d,** $f_0 \geq 220$ **Hz)**	**0.03**	**1.18**×	**0.1**%	**1.74**×	**240** Hz	**1.62**×	**96.5**%
9. Sim Stress: Orchard Wind Gusts (45 km/h, 64-Frame Filter)	**0.03**	**1.16**×	**0.2**%	**1.71**×	**239** Hz	**1.59**×	**98.4**% (`Rej=1`)
10. Sim Stress: Post-Baffle Cold Snap (32.8°C Rollback)	**0.04**	**1.14**×	**0.4**%	**1.64**×	**236** Hz	**1.52**×	**94.2**% (`Abort=3`)

4 *In Silico* Simulation Audit of 8 Apiary Practices and Edge Stress Tests

Simulation Delineation: All 14-day colony trajectories and stress-test metrics in Table 3 are ***in silico* ODE computational simulations** parameterized from published respirometry and toxicology literature [1, 14, 2, 6, 7] to approximate holobiont dynamics and design prospective *in vitro* and safe dietary patty experiments:

1. **Simulated Failure of Unphased Feeding & Antibiotics (Rows 1–5):** Plain `HFCS` lacks *p*-coumaric acid, leaving simulated `CYP9Q1–3` uninduced ($S_{Q_o} = 0.72$, simulated homing = 18.0%). Prophylactic `oxytetracycline` ablates simulated *S. alvi* (0.04×), elevating gut lumenal O_2 and dropping simulated survival to 12.4%. Unphased Day-0 pollen substitute or Day-0 `CoQ10` pushes metabolic flux through 85%-blocked Complex III (`1SQB`), predicting only 31.2–38.4% survival. Formic acid flash fumigation adds Complex IV cytochrome *c* oxidase inhibition on top of Complex III blockade (21.5% simulated survival).
2. **Simulated Orchard Wind-Gust Noise Immunity (Row 9):** Under simulated 45 km/h orchard wind gusts and hive knocks (180–260 Hz acoustic broadband transients), a single-sample 220 Hz tripwire triggers premature Stage-2 `Bay-B` actuation in 68.4% of simulated runs. Our 64-frame rolling persistence buffer (`128 B` DTCM, Eq. (3)) achieves 0.0% false triggers (`WindGust_Rej=1`, 98.4% simulated homing survival).
3. **Simulated Post-Baffle Cold-Snap Rollback (Row 10):** When a simulated overnight orchard frost drops brood-nest temperature to 32.8°C and induces a 300 Hz worker distress hiss after `Bay-B` actuation, a one-way feeder model predicts 28.5% survival. Executing `BCR_GATE_STAGE2_ROLLBACK_ABORT` (3) snaps the simulated bistable solenoid back to the `Bay-A` *p*-coumaric detox patty (`GPIO_BAFFLE=1`) and restores the 32-byte DTCM breadcrumb, predicting 94.2% Day-14 survival.

5 Prospective *In Vitro* Respirometry and Safe Dietary Phytochemical Protocol (Prediction P4)

Because *Apis mellifera* requires zero vertebrate animal (IACUC) or human IRB approval (TRL 5 → 7 in 14 days), and utilizes exclusively **safe, non-harmful natural floral phytochemicals and dietary supplements** (*p*-coumaric acid monomer from natural pollen coats, spermidine, `Ubiquinol CoQ10`, and royal-jelly `10-HDA`), we specify a two-tier prospective validation protocol to test our *in silico* simulations:

1. **Tier 1 — 48-Hour *In Vitro* Permeabilized Flight-Muscle Respirometry:** Dissect dorsal longitudinal

flight muscle from newly emerged worker bees, permeabilize fibers with 50 μg/mL saponin in MiR05 buffer, and measure high-resolution respirometry in an `Oroboros Oxygraph-2k` at 35°C with simultaneous fluorometric Amplex Red H_2O_2 detection. Challenge fibers with 2.0 μM azoxystrobin (`PDB: 1SQB`) and compare State-3 O_2 flux and H_2O_2/O_2 flux ratios between fibers from bees pre-fed 500 μM *p*-coumaric acid for 48 h vs. acute unphased `Ubiquinol` addition.

2. **Tier 2 — 14-Day Incubator Hoarding-Cage & RFID Hive Assay ($N = 50$ Bees/Cage $\times 4$ Arms, 34.5°C, 65% RH):** Expose bees orally to 5 ppm azoxystrobin in 50% sucrose. Compare Arm 1 (Plain Sucrose/`HFCS`), Arm 2 (Day-0 Pollen Patty + `Ubiquinol`), Arm 3 (*p*-Coumaric Acid 500 μM + *S. alvi* 10^7 CFU/mL continuous), and Arm 4 (Closed-Loop **Apis-BCR** gated by the top-bar MEMS microphone at $f_0 \geq 220.0$ Hz on Day 4.2 into 200 μM `Ubiquinol` + 1.5 mM `10-HDA`). The hypothesis is falsified unless Arm 4 achieves $\geq 2.0\times$ higher Day-14 survival and 1 km RFID homing return rate than Arms 1 and 2 ($p < 0.0001$, log-rank test).

A Complete Holobiont ODE System and Auxiliary Readout Equations

Let $\mathbf{z}_{\text{Apis}}(t) = [W, D, S_{Q_o}, C_{\text{CYP}}, S_{\text{alvi}}, V_g, T_{\text{brood}}, H_{\text{surv}}]^\top \in \mathbb{R}_+^8$ denote the eight dynamic ODE states and $(\text{ATP}_{\text{th}}(t), f_0(t))$ the two coupled algebraic physiological readouts (forming the 10-variable colony trajectory $\mathbf{x}_{\text{Apis}}(t)$ over $t \in [0, 14\text{ d}]$). With heteroplasmy $h(t) = D(t)/[W(t) + D(t)]$, the complete eight-state ODE system is:

$$\frac{dW}{dt} = k_{\text{bio}}(u_2, V_g)\, W \left(1 - \frac{W+D}{K}\right) - k_{\text{mito},W}\, W - k_{\text{dmg}} \max\big(0,\ \text{EP}_{\text{Apis}} - \theta_{\text{GSH}}\big) W, \tag{A.1}$$

$$\frac{dD}{dt} = k_{\text{rep,adv}}\, k_{\text{bio}}(u_2, V_g)\, D \left(1 - \frac{W+D}{K}\right) - k_{\text{mito},D}(u_1)\, D + k_{\text{dmg}} \max\big(0,\ \text{EP}_{\text{Apis}} - \theta_{\text{GSH}}\big) W, \tag{A.2}$$

$$\frac{dS_{Q_o}}{dt} = J_{\text{exposure}} - k_{\text{detox}}\, C_{\text{CYP}}\, S_{Q_o}, \tag{A.3}$$

$$\frac{dC_{\text{CYP}}}{dt} = k_{\text{CncC}} \left(u_{\text{pCoum}} - C_{\text{CYP}}\right), \tag{A.4}$$

$$\frac{dS_{\text{alvi}}}{dt} = r_{\text{alvi}}\, S_{\text{alvi}} \left(1 - \frac{S_{\text{alvi}}}{K_{\text{gut}}}\right) (1 - \alpha_{\text{Tet}}\, u_{\text{Tet}}) + u_{\text{probiotic}}, \tag{A.5}$$

$$\frac{dV_g}{dt} = k_{V_g,\text{syn}}\, (1 + 2.2\, u_{\text{10HDA}})\, (1 - h) - k_{V_g,\text{ox}}\, \text{EP}_{\text{Apis}}\, V_g, \tag{A.6}$$

$$\frac{dT_{\text{brood}}}{dt} = k_{\text{therm}} \left(\text{ATP}_{\text{th}}\, (T_{\text{set}} - T_{\text{brood}}) - \gamma_{\text{amb}}\, (T_{\text{brood}} - T_{\text{amb}}(t))\right), \tag{A.7}$$

$$\frac{dH_{\text{surv}}}{dt} = -k_{\text{mort}} \max\Big(0,\ 1.10 - \text{ATP}_{\text{th}} + 0.5 \max\big(0,\ \text{EP}_{\text{Apis}} - \theta_{\text{GSH}}\big)\Big) H_{\text{surv}}. \tag{A.8}$$

The stage-modulated rate laws, gut oxygen-dysbiosis penalty $\Phi_{O_2}(S_{\text{alvi}})$, energetic stress potential EP_{Apis}, thoracic ATP capacity ATP_{th}, and wingbeat fundamental frequency $f_0(t)$ are given explicitly by:

$$k_{\text{bio}}(u_2, V_g) = k_{\text{bio},0} + \left(k_{\text{bio,max}} - k_{\text{bio},0}\right) u_2 \left(\frac{V_g}{K_{V_g} + V_g}\right), \tag{A.9}$$

$$k_{\text{mito},D}(u_1) = k_{\text{mito},W} + \left(k_{\text{mito},D,\text{max}} - k_{\text{mito},W}\right) u_1, \tag{A.10}$$

$$\Phi_{O_2}(S_{\text{alvi}}) = 1 + \beta_{O_2} \max\left(0,\ 1 - \frac{S_{\text{alvi}}}{K_{\text{gut}}}\right), \tag{A.11}$$

$$\text{EP}_{\text{Apis}} = \left(\frac{1 + 2.4\, S_{Q_o} + 1.8\, h + \alpha_{\text{formic}}\, u_{\text{formic}}}{(W + 0.12\, D)(1 + 0.35\, V_g)}\right) \left(\frac{f_0}{100\ \text{Hz}}\right)^2 \Phi_{O_2}(S_{\text{alvi}}), \tag{A.12}$$

$$\text{ATP}_{\text{th}} = \frac{(W + 0.12\, D)(1 + 0.45\, u_{\text{Ubiq}})}{1 + 1.85\, S_{Q_o} + 0.90\, u_{\text{formic}}}, \tag{A.13}$$

$$f_0(t) = 100\ \text{Hz} + 140\ \text{Hz} \cdot \min\left(1.15,\ \frac{\text{ATP}_{\text{th}}(t)}{1.35}\right). \tag{A.14}$$

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